

Transformations of continuous random variables

▮ You are given the PDF of X .

▮ Let $Y = g(X)$.

▮ The PDF of Y can be obtained via three steps :

① Find $F_X(x) = \int_{-\infty}^x f_X(x) dx$

② Find $F_Y(y) = P(Y \leq y)$
 $= P(g(X) \leq y)$

$$= \begin{cases} P(X \leq g^{-1}(y)) , & \text{if } g \text{ is } \nearrow \\ P(X \geq g^{-1}(y)) , & \text{if } g \text{ is } \searrow \end{cases}$$

$$= \begin{cases} F_x(g^{-1}(y)) , & \text{if } g \text{ is } \nearrow \\ 1 - F_x(g^{-1}(y)) , & \text{if } g \text{ is } \searrow \end{cases}$$

$$\textcircled{3} \quad f_Y(y) = \frac{d}{dy} F_Y(y)$$

$$= \begin{cases} f_x(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) , & g \text{ is } \nearrow \\ -f_x(g^{-1}(y)) \cdot \frac{d}{dy} g^{-1}(y) , & g \text{ is } \searrow \end{cases}$$

$$= \boxed{f_x(g^{-1}(y)) \cdot |J|}$$

only works
for monotonic g

where $J = \frac{d}{dy} g^{-1}(y) = \frac{d \text{ "old" }}{d \text{ "new" }}$

Ex. Let X have PDF

$$f_X(x) = \begin{cases} 6x(1-x), & x \in (0,1) \\ 0, & \text{o.w.} \end{cases}$$

Let $Y = g(X) = X^3$. Find $f_Y(y)$.

$\therefore g(x)$ is $\nearrow$ for $x \in (0,1)$

$$\therefore f_Y(y) = f_X(g^{-1}(y)) \cdot |J|$$

where $g^{-1}(y) = y^{\frac{1}{3}}$ and

$$J = \frac{d}{dy} g^{-1}(y) = \frac{1}{3} y^{-\frac{2}{3}}$$

$$f_Y(y) = 6 \cdot y^{\frac{1}{3}} \cdot (1 - y^{\frac{1}{3}}) \cdot \frac{1}{3} y^{-\frac{2}{3}}, \quad y \in (0,1)$$

▮ Ex. Let X have PDF

$$f_X(x) = \begin{cases} \frac{1}{2} & , x \in (-1, 1) \\ 0 & , \text{o.w.} \end{cases}$$

Let $Y = X^2$. Find $f_Y(y)$.

NOTE: $g(x) = x^2$ is NOT monotonic over $(-1, 1)$

$$\text{Step ① } F_X(x) = \int_{-1}^x \frac{1}{2} dx = \frac{1}{2}(x+1)$$

$$\text{Step ② } F_Y(y) = P(X^2 \leq y)$$

$$= P(-\sqrt{y} \leq X \leq \sqrt{y})$$

$$= F_X(\sqrt{y}) - F_X(-\sqrt{y})$$

$$= \frac{1}{2}(\sqrt{y}+1) - \frac{1}{2}(-\sqrt{y}+1) = \sqrt{y}.$$

Step ③ $f_Y(y) = \frac{d}{dy} F_Y(y)$

$$= \frac{1}{2} y^{-\frac{1}{2}}, \quad y \in (0, 1).$$